ADVANCED NETWORKS AND SECURITY / HALMSTAD UNIVERSITY

From 5G to 6G

Emerging cybersecurity challenges and implications for industry

Literature review proposal 9 November 2026

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Research question

How might proposed 6G capabilities change the way cybersecurity teams secure, test, and monitor wireless networks compared with 5G?

5G baseline

Published 5G security architecture and procedures.

6G scope

Capabilities described in the ITU's IMT-2030 framework and related research.

Sources [1], [2], [4]

Three areas for comparison

6G capability	Security question for practitioners
AI in networks	How could teams detect manipulated model inputs and review automated decisions?
Wireless sensing	Who can request sensing and who can access the results?
IoT at greater scale	How could operators revoke credentials across larger fleets?

ISAC combines wireless communication with sensing.

Capability and risk context [1], [3], [5]. Security questions are proposed by our group.

Review method

01 Select sources

Use standards documents and peer-reviewed research relevant to the three areas.

02 Extract claims

Record each claim, its source, and whether it is a specification, proposal, or research finding.

03 Compare

Link each capability to a potential change in security practice and explain what remains uncertain.

Method **Focused literature review**

Analysis of published documents, with the same comparison criteria for each area.

Source set [1]–[5]

Worked example for IoT device security

Evidence status	What we can say
Specified in 5G	5G specifies subscription and network authentication. [2]
6G framework objective	IMT-2030 builds on the 5G scenario for connecting large numbers of devices. [1]
Our inference	Operators may need to assess whether credential management and revocation can scale to larger fleets.

The impact depends on network design and deployment choices.

5G security requirements [2, Section 5.1.2]. IMT-2030 usage scenarios [1].

Proposed contribution

A comparison matrix that connects 6G research to the work cybersecurity teams do.

- Identify which existing 5G security practices provide a baseline.
- Show where a proposed capability raises a new testing or monitoring question.
- State what evidence would help assess the potential impact in practice.

The review will separate conclusions supported by published documents from questions for future deployments.

Core references

1. **ITU-R (2023).** Recommendation M.2160-0, *Framework and overall objectives of the future development of IMT for 2030 and beyond*.

2. **3GPP (2022).** TS 33.501 v15.15.0, *Security architecture and procedures for 5G System*.

3. **ETSI (2026).** GR ISC 004, *Integrated Sensing And Communications (ISAC). Security, Privacy, Trustworthiness and Sustainability*.

4. **Porambage et al. (2021).** *The Roadmap to 6G Security and Privacy*. IEEE Open Journal of the Communications Society.

5. **De Alwis et al. (2026).** *Federated Learning for 6G Security: A Survey on Threats, Solutions, and Research Directions*. IEEE Communications Surveys & Tutorials.